AGOS

Flood Alert & Community Resilience Platform

Project Proposal · Research Foundation · System Design Documentation

Document Type	Version	Context	Status
Full Project Documentation	4.0 — March 2026	2-Day MVP Sprint + Phased Roadmap	v4.0 — Judge-Aligned: Regeneration Layer, Digital Inclusion, Sustainability, PhilAWARE Positioning, Trash Accountability Loop

From a Flood Survivor's Perspective

"Every time the rain gets heavy, I don't sleep. I check my phone, I look out the window — I want to know how much time I have to get my family out. The apps I've tried just show a map or push a generic alert. What I really need is something that tells me: the water on YOUR street is rising. Can you leave in the next 30 minutes? And after it floods — where do I get food, where do I register for aid? And after everything is over — will my street flood less next year? Nobody gives us that." — This documentation is built around that human need.

SECTION 1 — EXECUTIVE SUMMARY

AGOS (Filipino: ‘flow’ or ‘current’) is a mobile-first flood alert and community resilience application designed for flood-prone communities in the Philippines. Unlike existing alert systems that deliver generic warnings, AGOS is built on the lived experience of people who repeatedly face floods — providing hyper-local, actionable intelligence before, during, and after a flood event.

AGOS operates across three pillars that together form a complete flood resilience cycle:

- MITIGATION — Trash & Pollution Reporter, Household Preparedness Kit, drainage hotspot mapping that addresses the root cause of urban flooding before it happens.
- RESPONSE — Real-time Google Flood Forecasting alerts, Community-Informed Route Advisor, Evacuation Decision Engine, offline fallback, SMS continuity, multilingual voice interface covering all Philippine regional languages.
- REGENERATION — Post-Flood Relief Tracker, Trash Report Lifecycle with LGU accountability, Community Resilience Dashboard (pollution trend tracking), drainage rehabilitation milestones, and a sustainability model for long-term platform operation.

The application also directly addresses the digital divide: a Digital Inclusion Bridge ensures that AGOS serves households without smartphones through Flood Buddy community relay volunteers, barangay-level shared devices, and bulk SMS delivery to non-app users. AGOS is explicitly designed to contribute to the UNDRR Early Warnings for All (EW4All) initiative — the UN’s target of universal early warning coverage by 2027.

Target Users	Primary Market	Core Technology	MVP Timeline
Flood-prone households, LGUs, first responders, and non-smartphone users via Flood Buddy relay	Metro Manila (PRIMARY); future: the whole Philippines	React Native, Firebase, Google Flood Forecasting API (PRIMARY), Multilingual STT, Google Places API	48-hour sprint + 30-day v1.0 roadmap

SECTION 2 — PROBLEM STATEMENT & USER PAIN POINTS

2.1 The Scale of the Problem

Flood events have increased fourfold since 1980. Between 2011 and 2020, floods killed approximately 45,000 individuals globally, with 73% of casualties in lower-income nations (EASAC, 2018; Guha-Sapir, 2020). The Philippines experiences an average of 20 typhoons per year; Metro Manila and surrounding provinces suffer chronic urban flooding, displacing hundreds of thousands annually.

A critical but under-documented driver of this flooding is solid waste. The Department of Public Works and Highways (DPWH) discloses that 70% of Metro Manila's drainage systems are clogged with garbage — predominantly plastic sachets and single-use packaging. Research confirms drainage capacity drops to 3.6% at 40% clogging and 1.4% at 60% clogging (IJSRED, 2025). Yoshioka, Era, & Sasaki (2021) document that climate disaster risk and waste management must be addressed as a single integrated problem in Philippine communities — not as separate policy silos. AGOS's Trash Reporter is built on this principle.

2.2 Why Existing Systems Fall Short

A comprehensive review of 77 flood apps in the Google Play Store (World Mobile Technology Review, Urban Science, MDPI, 2025) found most fail on four critical dimensions:

- Generic, city-wide alerts with no street-level specificity
- Over-reliance on outdated government data. DOST PAGASA data is unreliable for real-time use. AGOS does not use PAGASA — Google Flood Forecasting API (free) is the sole real-time data source.
- No post-disaster continuity — apps go silent after the storm passes and provide no regeneration or recovery support
- No offline fallback — useless precisely when connectivity drops during a typhoon; low community engagement
- No digital inclusion strategy — non-smartphone users, who are disproportionately the most flood-vulnerable, are entirely unserved

Real Case — Typhoon Gaemi (Carina), Metro Manila, July 2024:

Jeepney driver Allan Figueroa's vehicle was destroyed when his neighborhood flooded unexpectedly. Forecasts told him rain was coming. They did not tell him his specific street would flood within the hour. And no app told him his drainage canal was already choked with plastic. After the flood, no app told him where to get food, how to register for aid, or how to report the blocked estero so it doesn't happen again next year. AGOS is built so Allan gets all of that information — before, during, and after the flood.

2.3 Pain Points: What Flood Survivors Actually Need

What Survivors Actually Need	What Most Apps Currently Offer

Street-level flood depth updates — runs as background service; permission modal on every app open if revoked	Generic weather advisories and signal level numbers only
Clear evacuation decision with multilingual speech-to-text in any Philippine language	Static hazard maps based on historical data, not real-time
Community-informed safe route to nearest open evacuation center, with offline cached fallback	Government news feeds requiring active internet browsing
Post-flood: food distribution, relief registration, medical aid, one household one donation	Push notifications that stop after the storm passes
Offline fallback: cached instructions + SMS alerts + post-flood SMS updates	No offline mode — useless when connectivity drops mid-typhoon
Digital inclusion: non-smartphone households reached via Flood Buddy relay + bulk SMS	No multilingual support, no digital inclusion for non-smartphone users
Trash Reporter: photograph clogged drains with phone camera, track LGU cleanup accountability	No pollution or drainage-blockage reporting, no post-flood environmental recovery
Regeneration: pollution trend tracking, estero rehabilitation milestones, long-term resilience metrics	No regeneration layer — no environmental recovery, no community rebuilding support

SECTION 3 — REVIEW OF RELATED LITERATURE & EXISTING SYSTEMS

3.1 IoT-Based Early Warning Systems

Mohammed Siddique, Ahmed, & Husain (2023) confirm real-time automated measurement surpasses manual methods — especially critical given that flood peaks often occur before personnel can arrive. Prathaban et al. (2023) acknowledge that hardware deployment costs remain prohibitive at community scale in developing nations. This validates AGOS's software-first, crowdsourcing-supplemented approach.

3.2 Crowdsourcing as Flood Intelligence

See (2019) documents crowdsourced photograph analysis achieving 72–95% accuracy in street flood detection — comparable to automated sensors at a fraction of the cost. Samonte, Rozario, & Aranjuez (2017) developed a crowdsourced flood monitoring app for Philippine communities; their core principle — affected residents are the fastest and most accurate sensors — directly informs AGOS's reporting architecture.

A 2024 Reuters/Context report confirms NOAH (DOST) now solicits citizen flood photo submissions as mainstream government-endorsed data strategy. AGOS plans a Phase 2 NOAH partnership. Critically, AGOS addresses the crowdsource cold-start problem by seeding the flood map via Google Flood Forecasting API thresholds and MMDA EFCOS water-level data before community reports arrive.

3.3 Crowdsourced Pollution Reporting and the Flood–Waste Link

MacAfee, Ntelekos, & Kron (2024) establish that mismanaged plastic waste and urban flooding are deeply interlinked challenges in Southeast Asia. Yoshioka, Era, & Sasaki (2021) conducted participatory mapping across Philippine coastal communities, finding that waste pollution constitutes a direct component of flood vulnerability, and that national disaster risk policy and solid waste management policy must be integrated. AGOS's Trash Reporter — with its full report lifecycle, LGU accountability loop, and Community Resilience Dashboard — bridges this gap and transforms reporting into measurable environmental regeneration.

3.4 Impact-Based and Site-Specific Alert Mobile Applications

Meléndez-Landaverde & Sempere-Torres (2023) demonstrated that location-specific warnings with explicit self-protection action prompts substantially improve community preparedness. AGOS directly adapts this architecture for Philippine urban contexts. A Phase 2 partnership with Project NOAH (DOST) is planned for official data-sharing and co-branding.

3.5 AI and Machine Learning in Early Warning Systems

Toonstra, Pappenberger et al. (2025) found 55% of AI-EWS research addresses hazard prediction while only 1% addresses warning dissemination effectiveness. This gap — between generating good forecasts and reaching vulnerable people with actionable instructions — is precisely the space AGOS occupies. AGOS also aligns with the UNDRR Early Warnings for All (EW4All) four-pillar framework: (1) risk knowledge, (2) monitoring and warning service, (3) dissemination and communication, and (4) response capability. Each AGOS feature maps to one or more of these pillars.

3.6 Social Vulnerability and Metro Manila Flood Risk

Dulawan, Imamura, Amaguchi, & Ohara (2024) studied 1,169 households across Manila, Marikina, and Paranaque. They conclude that flood risk assessments must incorporate social vulnerability indicators — age, poverty, disability, gender — alongside physical hazard data. This validates AGOS's offline SMS fallback, large-text accessibility mode, Digital Inclusion Bridge for non-smartphone users, PhilSys-compatible ID verification, and Google Places API resource map. Post-flood support specifically targets socially vulnerable households through the one-household-one-donation policy and barangay co-verification protocol.

3.7 Google's Flood Forecasting — The Last-Mile Problem

Yale University's research team (2024) assessed Google's AI-based flood forecasting system across 319 village communities and 1.8 million people in South Asia. The study confirms the critical importance of community relay layers and offline fallbacks. This directly validates AGOS's use of Google Flood Forecasting API as its sole PRIMARY real-time data source, its Flood Buddy relay network for last-mile community coverage, and its SMS fallback. PAGASA is not integrated into AGOS due to outdated and unreliable data feeds.

3.8 PhilAWARE and the Institutional Landscape

The Office of Civil Defense deployed PhilAWARE in 2024 — an integrated multi-hazard monitoring and early warning platform serving institutional decision-makers at the DRRMO and OCD level. AGOS is not a competitor to PhilAWARE. The two platforms are complementary: PhilAWARE operates at the government-institutional tier; AGOS operates at the last-mile community tier, reaching individual households, non-smartphone users, and barangay coordinators with the granular, street-level information that no government platform currently provides. A Phase 2 integration is planned: AGOS community flood depth reports and pollution hotspot data would feed into PhilAWARE's institutional dashboard, strengthening both platforms.

SECTION 4 — WHAT MAKES AGOS DIFFERENT

AGOS is built around ten differentiators. No single existing Philippine or global flood application currently combines all of these:

#	Differentiator	How AGOS Addresses It
01	**Hyper-Local Precision**	Street-level flood depth mapping from crowdsourced reports + Google Flood Forecasting API (PRIMARY; free). PAGASA is not used. Background service with a permission modal ensures alerts are never silently dropped. Map pre-seeded via API thresholds + MMDA EFCOS water-level data to solve the cold-start problem.
02	**Community-Informed Route Advisor**	Renamed from 'AI Safe Route' for accuracy: Google Directions API call with crowd-report filtering routes users to the nearest open evacuation center. Offline fallback caches last 3 known-safe routes to nearby centers. 'Last known route' banner shown when offline.
03	**Offline Resilience**	When the internet is lost, AGOS serves pre-cached emergency instructions. Critical alerts delivered via Semaphore PH SMS (NDRRMC-style). Post-flood SMS updates inform users of food distribution, relief registration, and road passability status.
04	**Post-Flood Continuity + Donation**	Active support module: food distribution, digital relief registration, Google Places resource map. Donation system with name/address/government ID + PhilSys compatibility; one-household-one-donation policy with barangay co-verification for informal settlers.
05	**Multilingual Speech-to-Text**	AI-powered voice input and output supporting all Philippine regional languages. App detects user language and responds in kind — both interpreting commands and reading responses aloud. Hands-free use while evacuating.
06	**Trash & Pollution Reporter (+ Accountability Loop)**	Phone camera captures clogged drains, GPS-tagged and uploaded. Full report lifecycle: Submitted → LGU Notified → Dispatched → Resolved. 72-hour escalation if unacknowledged. Barangay Pollution Score creates accountability.
07	**Post-Flood Regeneration Layer**	Community Resilience Dashboard: long-term pollution heatmap trends showing whether drainage is improving. Estero rehabilitation milestones. Pollution report history feeds LGU and NGO environmental restoration planning. Closes the loop from reporting to measurable recovery.
08	**Digital Inclusion Bridge**	Non-smartphone users reached via: Flood Buddy relay volunteers (one volunteer serves 5–10 households), barangay shared-device deployment, bulk SMS to registered numbers regardless of smartphone ownership. Contributes to UNDRR EW4All universal coverage target.
09	**Complementary to PhilAWARE**	AGOS is the last-mile community interface that PhilAWARE does not provide. Phase 2: AGOS community reports feed into PhilAWARE's institutional dashboard. Positioned as a partner to the government EWS ecosystem, not a replacement.

#	Differentiator	How AGOS Addresses It
10	**Metro Manila Primary / Whole Philippines Vision**	MVP focused on Marikina, Malabon, Navotas, Pasig. Multilingual STT and SMS fallback are the national expansion enablers — ensuring AGOS works for any Filipino community regardless of language or connectivity.

SECTION 5 — CORE FEATURES & INNOVATIONS

5.1 Feature Overview

Feature	Description	Integration Difficulty
Real-Time Flood Alert	Push notifications (Watch / Warning / Danger) from Google Flood Forecasting API (PRIMARY). Barangay-specific. Runs as a background service; permission re-prompt modal on every app open if access was revoked. PAGASA is not used.	🟢 **Easy**
Street-Level Depth Map	Community flood depth reports (ankle/knee/waist/chest) as color-coded pins. Map pre-seeded by Google Flood API thresholds + MMDA EFCOS water-level data. 3+ community reports in 200 m confirm the event. Photo optional.	🟢 **Easy**
Evacuation Center Finder	Directory of active centers with amenities and GPS routing. Capacity display removed — data unreliable. Google Places API supplies nearby hospitals, shelters, barangay halls, pharmacies, + emergency hotlines (BFP, NDRRMC, PNP, local DRRMO).	🟢 **Easy**
Offline Fallback Mode	Pre-cached emergency instructions from device storage. Last-known evacuation center list cached via AsyncStorage. SMS fallback via Semaphore PH (NDRRMC-style). Post-flood SMS updates: food distribution, relief registration, and road passability.	🟡 **Medium**
Household Preparedness Kit	72-hour emergency kit checklist. Items tagged: Easy (Day 1–2), Moderate (Week 2), Complex (Phase 2). Go-bag guide. Gamified with completion badges.	🟢 **Easy**
Multilingual Speech-to-Text	AI-powered voice input/output supporting all Philippine regional languages. Detects user language, interprets intent, and responds aloud in the same language. Enables hands-free use. Hands-free evacuation prompts, flood reporting, and emergency queries.	🔴 **Hard**
Google Places API Resources	Hospitals, barangay halls, shelters, pharmacies, and evacuation centers within 5 km. Emergency hotlines (BFP, NDRRMC, PNP, DRRMO) displayed alongside location results.	🟢 **Easy**
Evacuation Decision Engine	Plain-language output: “Prepare to evacuate within 30 minutes.” Synthesizes the Google Flood API advisory, the nearby depth cluster, and the household floor level. Rule-based v1; ML in Phase 2.	🟡 **Medium**
Community-Informed Route Advisor	Google Directions API with crowd-report filtering to the nearest open evacuation center. Last 3 safe routes cached	🔴 **Hard**

Feature	Description	Integration Difficulty
	offline. 'Last known route' banner shown when offline. Road passability crowd-reports sync on restore.	
Trash & Pollution Reporter	Phone camera captures trash/blocked drains; GPS-tagged, waste type selected, uploaded to Firebase. Orange pins on the map. 3+ pins in 200 m auto-alerts LGU. Full report lifecycle + 72-hr escalation + Barangay Pollution Score.	🟡 **Medium**
Digital Inclusion Bridge	Flood Buddy volunteers relay alerts to households without smartphones (1 volunteer: 5–10 homes). Bulk SMS alerts to all registered barangay numbers. Barangay shared-device deployment. Contributes to UNDRR EW4All universal coverage.	🟡 **Medium**
Post-Flood Regeneration Dashboard	Community Resilience Dashboard: long-term pollution heatmap trends per barangay. Estero rehabilitation milestones updated by LGU. Pollution history report for NGO/LGU environmental restoration planning.	🔲 **Phase 2**
Post-Flood Relief Tracker	Food distribution updates, relief registration, and medical aid locator via Google Places. SMS updates (NDRRMC-style) for offline users.	🔲 **Phase 2**
Donation System	Name/email/address + government ID (PhilSys compatible) for verification. One household, one donation. Barangay co-verification for informal settlers. Payrex (testing) → digital banking partners (production).	🔲 **Phase 2**
Flood Buddy Network	Volunteers (barangay health workers, Pantawid coordinators) register availability. Act as community relay and first reporters during flood events. Linked to the route advisor and the evacuation engine.	🔲 **Phase 2**
LGU Admin Portal	LGU coordinators: push alerts, update center info, view flood + pollution dashboard, broadcast post-flood SMS, mark trash reports resolved, manage barangay pollution scores.	🔲 **Phase 2**
PhilAWARE Integration	AGOS community flood depth + pollution hotspot data feeds into PhilAWARE's institutional dashboard. Positioned as a government ecosystem partner.	🔲 **Phase 2**
Media/NGO API Access	Verified real-time flood, pollution, and community data feed for media and NGO relief coordination.	🔲 **Phase 2**

5.2 Background Service & Permission Re-Prompt

The AGOS app runs as a background service to ensure flood alerts are always delivered even when the app is not actively open. On Android and iOS, background execution permissions can be revoked by the OS after extended inactivity or by the user.

- On every app open, the system checks whether background location and notification permissions are still active.

- If any permission has been revoked — regardless of how long ago — a modal prompt is immediately displayed requesting re-authorization.
- This modal cannot be dismissed without either granting permission or explicitly acknowledging reduced safety functionality.
- This ensures users who have not opened the app for extended periods are immediately prompted to restore full alert capability.

5.3 Multilingual Speech-to-Text with AI Language Detection

A significant proportion of Filipinos do not speak Tagalog or English as their primary language. AGOS implements a multilingual voice interface ensuring accessibility across all Philippine regional languages.

- User activates voice input on any screen (evacuation prompt, flood report, emergency question).
- AI-powered speech recognition with Philippine language model support: Cebuano, Ilocano, Hiligaynon, Waray, Kapampangan, Bikol, and others.
- Spoken input is transcribed and interpreted to determine intent (e.g., “Is it safe to go to Barangay X?” in Cebuano).
- Intent is translated to a structured query, processed against flood data, and a response is generated.
- Response is delivered both as text and as text-to-speech audio in the same language the user spoke.
- Enables hands-free use while carrying children, go-bags, or navigating during an emergency.

5.4 Trash & Pollution Reporter — Full Report Lifecycle

The Trash Reporter is not just a reporting tool — it is a community accountability and environmental regeneration mechanism. Every report has a lifecycle:

- User opens Report screen and taps to activate the in-app camera (requires camera + location permission on Android and iOS).
- User photographs trash accumulation, clogged drains, or blocked waterways. GPS auto-tags the location. Waste type selected: plastic / debris / silt / other.
- Report uploaded to Firebase Firestore; photo to Firebase Storage. Orange pin appears on the live map immediately.
- When 3+ pins cluster within 200 m, LGU sanitation admin receives an automatic hotspot alert in the Admin Portal.
- Report lifecycle screen tracks progress: Submitted → LGU Notified → Dispatched → Resolved. Original reporter receives a notification when their report is actioned.
- If a report is not acknowledged within 72 hours, it auto-escalates to barangay captain level in the Admin Portal.
- Each barangay has a public Barangay Pollution Score showing average resolution time and number of unresolved reports. This creates positive accountability pressure between LGUs.
- Aggregate resolved reports over time populate the Community Resilience Dashboard: a long-term map layer showing which areas are getting cleaner and which remain at risk.

♻️ This Is the Regeneration Loop

A user reports a clogged estero before typhoon season. The LGU is alerted, dispatches a cleanup crew within 48 hours, and marks the report resolved. The user sees their report led to action. The Community Resilience Dashboard shows that Barangay X's pollution hotspot count dropped by 40% over the past 3 months. That data is used by the LGU's sanitation office to plan the next drainage rehabilitation project. The estero that flooded Allan's street last year floods less this year. This is regeneration — not just response.

5.5 Post-Flood Relief & Donation System

Post-Flood Relief Updates

- Food distribution point locations and schedules — updated in real time by LGU admins
- Digital relief registration form for affected households
- Medical aid locator — powered by Google Places API (hospitals, medical missions, barangay health centers)
- Offline users receive SMS updates in NDRRMC-style format: brief, structured text messages with distribution locations and registration instructions

Donation System Design

- Donor registration requires: full name, email, home address, and upload of a valid government-issued ID. PhilSys (Philippine National ID) is the primary ID option — over 70 million Filipinos enrolled as of 2024, including many informal settlers.
- For informal settlers without a formal address: a Flood Buddy volunteer or barangay health worker can co-verify identity using a barangay-issued certificate, bypassing address-matching requirements.
- One household, one donation policy: enforced by address matching + barangay household registration code for shared-address situations (e.g., compound households).
- Payment processing: Payrex sandbox for development/testing. Production deployment integrates with established Philippine digital banking partners.
- All transactions are logged and auditable by system admins.

5.6 Offline Fallback: Emergency Instructions & SMS Continuity

- Pre-cached emergency instructions served from device storage: what to do when flooding starts, when to evacuate, how to signal for help, household safety protocols.
- Last-known evacuation center list and nearest emergency locations cached via AsyncStorage.
- Community-Informed Route Advisor caches the last 3 known-safe routes to nearby centers. Shown offline with a 'last known route' banner.
- SMS fallback via Semaphore PH: critical flood warnings and post-flood updates delivered to users without internet connectivity.
- Post-flood SMS updates continue to inform offline users: food distribution schedules, medical aid availability, relief registration deadlines, road passability.

Note on Evacuation Center Capacity

AGOS v4.0 does not display capacity availability in the Evacuation Center Finder. Reliable real-time capacity data cannot be guaranteed without direct LGU integration and consistent data entry. Displaying inaccurate capacity may cause people to bypass open centers or overload others. Capacity tracking will be re-evaluated for Phase 2 with proper LGU data partnerships.

5.7 Digital Inclusion Bridge — Reaching Non-Smartphone Users

The people most at risk from flooding in Metro Manila are disproportionately from low-income households, where smartphone ownership and internet access are limited. AGOS is explicitly designed to reach these users — not just those who already have a smartphone.

- Flood Buddy Relay: registered volunteers in each barangay serve as human relay points. One volunteer with the AGOS app alerts the 5–10 surrounding households that do not have smartphones — verbally, by knock, or by forwarded SMS.
- Bulk SMS Broadcasting: critical flood alerts and post-flood updates are sent via SMS to all registered phone numbers in a barangay — including non-smartphone users who registered only a mobile number. No app download required.
- Barangay Shared Device: AGOS proposes deployment of one AGOS-enabled tablet or device at every barangay hall as a public access point. LGU admins and walk-in residents can access the platform without a personal smartphone.
- UNDRR EW4All Alignment: These three mechanisms together contribute to the UN's Early Warnings for All target — ensuring that universal early warning coverage is not just a goal but an operational design principle of AGOS.

SECTION 6 — TARGET MARKET & USER PERSONAS

6.1 Primary Target Market

Metro Manila — PRIMARY PILOT: Marikina, Malabon, Navotas, and parts of Pasig. Chronic urban flooding from the Marikina River and Laguna Lake systems. Highest smartphone penetration. Largest user base potential. All MVP testing, LGU partnerships, and user recruitment concentrated here.

Future expansion target: the whole Philippines. Multilingual STT and SMS fallback are the national expansion enablers. The Digital Inclusion Bridge ensures AGOS is accessible to any Filipino community regardless of language or connectivity.

6.2 Secondary Markets

- LGU DRRMOs — Admin portal users for evacuation management, alert broadcasting, pollution response, and Community Resilience Dashboard monitoring
- Schools and universities in flood-prone areas — Institutional alert subscriptions
- Community organizations (barangay health workers, Pantawid coordinators) — Flood Buddy Network activators
- Media outlets and NGOs — API access for verified real-time flood, pollution, and regeneration data
- PhilAWARE / OCD — Phase 2 data-sharing partnership for institutional dashboard integration

6.3 User Personas

Persona 1: Maria, 42 — Marikina Homeowner (Survivor)

Maria has experienced flooding 7 times in the past decade. Before typhoon season, she photographs the choked estero behind her street and files a Trash Report in Tagalog. She checks the Barangay Pollution Score — it improved from last year. When the Google Flood Forecasting API triggers a Warning alert, the Evacuation Decision Engine tells her in Tagalog: “Handa na kayo umalis sa loob ng 30 minuto.” She says “navégahin mo na” and follows the cached safe route to the evacuation center. When connectivity drops, the app serves her cached emergency instructions. After the flood, she receives an SMS with the food distribution point address and registers for relief digitally. The donation system processes one donation for her household. Three weeks later, she checks the Community Resilience Dashboard and sees the LGU resolved her estero report.

Persona 2: Noel, 35 — Barangay Emergency Coordinator, Navotas

Noel currently coordinates evacuations manually via Facebook and phone calls. With AGOS Admin Portal he has a live dashboard: incoming flood depth reports, pollution hotspot alerts, evacuation center status. When three trash reports cluster near a pump station, he dispatches a cleanup crew proactively — before the storm. During the typhoon, he pushes verified official alerts to all registered residents in one tap. After the storm, he broadcasts a post-flood SMS with food distribution locations. He marks resolved trash reports in the portal, improving the barangay's Pollution Score. His AGOS community data feeds into PhilAWARE's institutional dashboard in Phase 2.

Persona 3: Lola Nena, 68 — Cebuano-Speaking Resident, Pasig

Lola Nena speaks only Cebuano and cannot read English or Tagalog. She opens AGOS and says: "Ligtas pa ba ang dalan padulong sa evacuation center?" The app detects Cebuano, processes her query against real-time road data, and reads the answer back to her in Cebuano. She receives flood alerts as voice announcements in her language. During offline periods, she receives SMS updates. She does not need to type or read anything.

Persona 4: Mang Totoy, 55 — No Smartphone, Informal Settler, Malabon

Mang Totoy has no smartphone and no internet. He lives in a compound with three other families sharing one informal address. During a typhoon, his neighbor Cora — a registered Flood Buddy volunteer — knocks on his door: "AGOS says the water is at knee level on the main road. We need to go now." He also receives a bulk SMS from the AGOS system directly to his feature phone. After the flood, he goes to the barangay hall's shared AGOS tablet to register for relief. His household receives one donation through the barangay co-verification protocol. AGOS reaches Mang Totoy because it was designed for him from the start.

SECTION 7 — SYSTEM ARCHITECTURE & TECHNICAL DESIGN

7.1 Five-Pipeline Data Flow

- Pipeline 1 — Automated Weather: Google Flood Forecasting API (PRIMARY AND ONLY) polled every 10 minutes. Threshold triggers auto-generate barangay-level alerts via Firebase Cloud Functions. PAGASA is not integrated.
- Pipeline 2 — Flood Bootstrapping: MMDA EFCOS water-level sensor data for Marikina River and major waterways pre-seeds the flood depth map before any community report arrives. Eliminates cold-start gap.
- Pipeline 3 — Community Crowdsource: Flood depth reports geotagged and cluster-validated (3+ in 200 m). Live on map within 2 minutes. Flood Buddy volunteers serve as designated first-reporters during active alerts.
- Pipeline 4 — Pollution Crowdsource: Trash and drain photo-reports geotagged as orange pins. 3+ pins in 200 m triggers LGU alert. Report lifecycle tracked to resolution. Aggregated into Community Resilience Dashboard.
- Pipeline 5 — Institutional: LGU/NDRRMC admin users push verified official alerts, update evacuation center info, broadcast post-flood SMS, and resolve pollution reports via Admin Portal. Phase 2: data feeds into PhilAWARE.

SECTION 8 — 2-DAY IMPLEMENTATION PLAN (SPRINT)

Team of 3: one full-stack developer, one React Native developer, one UI/UX designer doubling as QA. The 48-hour sprint delivers a functional, demonstrable MVP.

8.1 MVP Scope vs. Phase 2 Roadmap

✅ Delivered in 48-Hour MVP	🔲 Deferred to Phase 2 (Weeks 2–4)
• Google Flood Forecasting API alerts (barangay-level, PRIMARY) • Street-level depth map (pre-seeded by MMDA EFCOS + Google Flood API) • Evacuation center finder (no capacity — removed) • Background service + permission re-prompt modal on every app open • Offline fallback: cached instructions + last-3-routes + SMS alerts • Multilingual STT (Tagalog + 1 regional language in MVP) • Household preparedness kit • Rule-based Evacuation Decision Engine + voice input • Community-Informed Route Advisor (renamed from AI Safe Route) • Trash & Pollution Reporter with full report lifecycle + Barangay Pollution Score • Google Places API: hospitals, barangay halls, shelters, emergency hotlines • Digital Inclusion Bridge: Flood Buddy relay + bulk SMS to non-app users	• Full multilingual STT across all Philippine languages • Post-Flood Regeneration Dashboard + estero rehabilitation milestones • Post-Flood Relief Tracker (food, registration, medical) + SMS updates • Donation system with PhilSys verification + barangay co-verification • ML-based Evacuation Decision Engine with historical flood data • Flood Buddy Network (formal registration + management) • Full LGU Admin Portal with device verification • PhilAWARE data-sharing integration • Project NOAH partnership • Community Resilience Dashboard (long-term pollution trends) • Media/NGO API access • Gamification badges and household readiness score dashboard

SECTION 9 — POST-FLOOD REGENERATION FRAMEWORK

AGOS is designed to operate across all three pillars of flood resilience: mitigation before the flood, response during and immediately after, and regeneration — the long-term rebuilding of community safety and environmental health. Most flood apps stop at response. AGOS does not.

9.1 The Three-Pillar Resilience Cycle

🛡️ MITIGATION	⚡ RESPONSE	♻️ REGENERATION
• Trash & Pollution Reporter • Household Preparedness Kit • Drainage hotspot mapping • Pre-flood community reporting	• Real-time flood alerts • Evacuation Decision Engine • Community-Informed Route Advisor • Offline fallback + SMS alerts • Digital Inclusion Bridge • Multilingual STT	• Post-Flood Relief Tracker • Donation system • Report lifecycle + LGU accountability • Barangay Pollution Score • Community Resilience Dashboard • Estero rehabilitation milestones • PhilAWARE integration

9.2 Community Resilience Dashboard

The Community Resilience Dashboard is the long-term regeneration layer of AGOS. It aggregates pollution report data over months and years to produce:

- Per-barangay pollution heatmap trends: shows whether drainage conditions are improving, stable, or deteriorating over time
- Resolved vs. unresolved hotspot ratio per barangay: reveals which LGUs are responding to reports and which are not
- Seasonal flood correlation: links historical flood event data with pollution report density to identify which drainage blockages are most predictive of flooding
- Estero rehabilitation milestones: LGUs and NGOs can log major cleanup and rehabilitation actions, creating a public record of environmental recovery effort

This dashboard serves three audiences: residents who want to know if their neighborhood is getting safer, LGU sanitation offices planning drainage rehabilitation projects, and NGOs and government agencies allocating environmental restoration resources.

9.3 UNDRR Early Warnings for All (EW4All) Alignment

AGOS explicitly aligns with the UNDRR Early Warnings for All initiative — the UN's target of ensuring every person on Earth is protected by an early warning system by 2027. The four pillars of EW4All map directly to AGOS features:

EW4All Pillar	AGOS Implementation
(1) Risk Knowledge	Trash & Pollution Reporter + Community Resilience Dashboard identifies and tracks flood-risk factors at the street level before events occur.
(2) Monitoring & Warning Service	Google Flood Forecasting API (PRIMARY) + MMDA EFCOS water-level bootstrapping + community crowdsource depth reports provide multi-layer, real-time flood monitoring.
(3) Dissemination & Communication	Multilingual STT + FCM push alerts + bulk SMS + Flood Buddy relay ensures warnings reach every resident regardless of language, smartphone ownership, or connectivity.
(4) Response Capability	Evacuation Decision Engine, Community-Informed Route Advisor, offline fallback, Digital Inclusion Bridge, Post-Flood Relief Tracker, and the Donation System together form a complete response capability layer.

SECTION 10 — FUTURE PARTNERSHIPS, EXPANSION & SUSTAINABILITY

10.1 Project NOAH Partnership (Phase 2 Goal)

AGOS aims to formally partner with Project NOAH (DOST), which operates the most comprehensive scientific flood hazard mapping system in the Philippines. A Phase 2 NOAH partnership would enable official data-sharing, integration of NOAH's hydrological sensor network into AGOS flood depth maps, cross-promotion, and DOST-endorsed credibility.

Note on NOAH Partnership

The NOAH partnership is a future aspirational goal and is NOT part of the current MVP. Documentation acknowledges this intent to inform future stakeholder conversations and funding applications.

10.2 PhilAWARE Positioning and Integration

AGOS is the last-mile community interface that PhilAWARE does not provide. PhilAWARE serves institutional decision-makers at the DRRMO and OCD levels. AGOS serves individual households, non-smartphone users, and barangay coordinators at street-level granularity. The two platforms are complementary. Phase 2 integration: AGOS community flood depth reports and pollution hotspot data feed into PhilAWARE's institutional dashboard, strengthening both platforms and positioning AGOS as a formal partner in the government EWS ecosystem.

10.3 Sustainability & Business Model

AGOS is designed to be financially sustainable beyond the initial development phase. The sustainability model operates on four streams:

- LGU Admin Portal Subscription: Barangay and city-level LGUs subscribe to the Admin Portal (flood dashboard, alert broadcasting, pollution management, post-flood SMS). Priced to be accessible for municipal budgets, with a free tier for small barangays.
- NDRRMC/DICT/DOST Grant Pathway: AGOS qualifies for disaster resilience technology grants from DICT's e-Government Fund, DOST's Grants-in-Aid program, and NDRRMC's disaster preparedness budget. National rollout funding is expected from this channel.
- Google.org and AWS Disaster Response Programs: both offer free cloud credits for disaster relief and climate resilience technology. AGOS qualifies for these programs, significantly reducing operational cloud costs.
- NGO/Media API Access (B2B): Phase 2 verified API subscriptions for media organizations and NGOs provide a revenue stream that scales with the platform's community data value.

The Philippine Disaster Resilience Foundation (PDRF) is identified as a potential private-sector funding and partnership channel for operational scaling.

10.4 Geographic Expansion

Current Focus: Metro Manila — Marikina, Malabon, Navotas, Pasig. Highest chronic flood frequency and smartphone penetration.

Near-term Expansion: Cagayan Valley (high flood frequency, semi-rural), Pampanga & Bulacan (prolonged inundation, high post-flood support needs).

Long-term Vision: The whole Philippines. Multilingual STT and SMS fallback are the enablers of national expansion. The Digital Inclusion Bridge and Flood Buddy network model are designed to function effectively in areas with limited internet infrastructure.

10.5 Media Outlets and NGO API Access

Phase 2: AGOS will offer a verified API feed of real-time flood depth data, pollution hotspot data, community reports, and Community Resilience Dashboard metrics to media organizations and NGOs. Access requires organizational verification to maintain data integrity.

SECTION 11 — AI RELIABILITY, FAILURE MODES & MULTI-AGENT ARCHITECTURE

AGOS relies on AI for three critical functions during a live flood emergency: multilingual speech recognition, the Evacuation Decision Engine, and the Community-Informed Route Advisor. Each of these can produce wrong, ambiguous, or missing outputs. In a life-safety context, that is not acceptable without explicit safeguards. This section documents every identified failure mode, its mitigation, and the complete multi-agent architecture that governs how many AI systems run inside AGOS, what each one does, and what happens when any one of them breaks.

11.1 The Core Problem: What Happens When AI Gets It Wrong?

AI models in AGOS face three categories of failure:

Failure Type	What It Looks Like in AGOS	Why It Is Dangerous
Wrong Output	The Evacuation Decision Engine says 'Stay in place' even though the flood is already at waist level near the user.	**User trusts the AI and does not evacuate. This is a life-threatening error.**
No Output	The STT agent fails silently during the typhoon. User speaks but receives no response.	**User assumes the app is processing. Critical seconds were lost during the evacuation window.**
Dangerous Output	Route Advisor sends the user along a road that the community reports as flooded, even though reports have not yet reached the 3-report threshold.	**User physically enters floodwater following the app's direction.**
Overconfident Output	Decision Engine gives a firm recommendation without flagging that the underlying flood data is 20 minutes old.	**User acts on outdated information presented as current.**
Wrong Language Output	Language Agent detects Tagalog, but the user is speaking Hiligaynon. Emergency instruction is read out in the wrong language.	**The user cannot understand the instruction and cannot act on it.**

> ⚠️ **The Non-Negotiable Safety Rule**
> No AI output in AGOS ever triggers an irreversible action without user confirmation. The Evacuation Decision Engine recommends — it does not force. The Route Advisor suggests — it does not lock the screen. The STT system shows what it heard before acting on it. Every AI output that affects safety is labelled with its data source and timestamp. In a life-safety context, AI is a decision-support tool, not a decision-maker. Humans always have the final action.

11.2 Per-Component Failure Mode & Mitigation

AI Component	Failure Mode	What Goes Wrong	Mitigation & Fallback
Multilingual STT + Language Detection	**Wrong language detected**	App interprets Cebuano as Tagalog. Emergency instruction was delivered in the wrong language. The user cannot understand.	Confidence score threshold: if language detection confidence < 80%, the app shows the detected language with a one-tap 'Change language' prompt before responding. The user is never silently given wrong-language output.
Multilingual STT + Language Detection	**Noisy environment transcription error**	Heavy rain and wind cause mis-transcription. The user's query is garbled before it reaches the intent system.	Transcription is shown on screen before processing. User sees exactly what the system heard and can tap 'That's wrong — re-record' or switch to text input at any point.
Multilingual STT + Language Detection	**Unsupported dialect**	User speaks Yakan, Ivatan, or Tausug. STT returns zero confidence or error.	Auto-fallback to text input mode. On-screen message: 'Voice input could not understand your language. Please type below.' Text input is always available regardless of STT status.
Evacuation Decision Engine	**Wrong recommendation (stay vs. evacuate)**	The engine says 'Stay in place' when the flood is rising. Caused by stale API data or a wrong barangay geolocation match.	All recommendations show their data sources and timestamp: 'Based on Google Flood API (updated 8 min ago) + 4 community reports near your street.' Data older than 15 min triggers a red banner: 'Warning: flood data may be outdated.' LGU-pushed alerts always override engine output.
Evacuation Decision Engine	**Conflicting inputs, no clear output**	Google Flood API says Watch level. 8 community reports say chest-level water. The engine cannot resolve.	Conflict resolution rule: 5+ community reports within 200 m of the user override the API level. Output surfaces both signals: 'API: Watch level. Community reports near you: chest-level water. Treating this as a Danger level.' Always errs toward evacuation.
Community-Informed Route Advisor	**Route through a flooded road**	Crowd reports have not yet reached the 3-report threshold for a flooded segment. The advisor does not know it is impassable.	All route outputs include a live caution banner: 'Road conditions may change. Watch for water.' Route is never presented as 'guaranteed safe' — only 'best available based on current reports.' If user reports a road impassable, their map updates immediately.

AI Component	Failure Mode	What Goes Wrong	Mitigation & Fallback
Community-Informed Route Advisor	**API unreachable (offline, typhoon)**	Google Directions API is down. No route can be computed. User left with nothing.	Last 3 successfully computed routes to nearest centers cached in AsyncStorage. Offline: cached route shown with banner: 'Last known safe route — conditions may have changed.' If no cached route exists: straight-line bearing and distance to nearest center displayed.
Community-Informed Route Advisor	**Destination center closed mid-route**	User is routed to a center that LGU closes after the route is loaded.	The LGU marks center is closed in the Admin Portal. AGOS pushes an in-app alert to all users en route: 'Your destination is now closed. Rerouting to [Center Name].' Re-route computed automatically.

11.3 Multi-Agent Architecture — How Many AI Agents Does AGOS Use?

AGOS uses 4 discrete AI agents. Each agent has a single defined responsibility, its own input/output contract, and an independent fallback. This isolation means one agent failing does not cascade into other agents failing — critical in a real-time flood emergency where partial functionality is far better than total failure.

Answer: AGOS uses 4 AI Agents

Agent 1: Speech & Language Agent | Agent 2: Query Intent Agent | Agent 3: Evacuation Decision Agent | Agent 4: Response Language Agent

Each agent is independent. Each has its own fallback. They run sequentially in a pipeline. If any one fails, that step falls back — the pipeline does not continue with bad data.

#	Agent	Responsibility	Input	Output	Fallback if Broken
1	**Speech & Language Agent**	Transcribe voice. Detect the user's language. Route to the correct language model.	*Audio stream from phone mic*	Transcribed text + language code + confidence score	Drops to text input. User types.
2	**Query Intent Agent**	Parse transcribed text into a structured system query. Determine what the user is asking: evacuation	*Text + language code (from Agent 1)*	Structured intent object: { intent, location, urgency }	Returns top 2 best-guess intents. User taps the correct one.

#	Agent	Responsibility	Input	Output	Fallback if Broken
		route, flood status, shelter, food, or emergency call.			
3	**Evacuation Decision Agent**	Synthesize flood data inputs into a single plain-language evacuation recommendation. Rule-based v1; ML Phase 2.	*Google Flood API level + community depth reports + household floor level*	Plain-language recommendation + data sources + timestamp + confidence flag	Surfaces all raw data inputs and asks user to decide. Errs toward evacuation if uncertain.
4	**Response Language Agent**	Translate system output into the user's detected language. Convert to text-to-speech audio.	*Response text + language code (from Agent 1)*	Audio (TTS) + on-screen text in the user's language	Displays text in Tagalog + English side by side. No audio.

11.4 Agent Interaction Pipeline

The four agents run sequentially during any voice-driven interaction. Each agent's output becomes the next agent's input. If any agent fails, the fallback activates at that exact step — bad data never propagates forward.

Voice Interaction Pipeline — Normal Flow

🎤 **User speaks → [Agent 1: Speech & Language] → Transcription + language code →**
[Agent 2: Query Intent] → Structured intent object →
[Agent 3: Evacuation Decision / System Logic] → Response text →
[Agent 4: Response Language] → 🔊 Audio + on-screen text in user's language

⚠️ **If any agent fails: fallback activates at that step only. Pipeline stops cleanly. User is never left with silence.**

11.5 Why 4 Separate Agents and Not 1 Large Model?

- Single point of failure avoided: if one model handled all four responsibilities and crashed during a typhoon, the entire voice interaction system would fail. With 4 agents, a broken STT still allows text input; a broken language agent still allows Tagalog/English output.
- Accountability: with 4 agents, if a recommendation is wrong, the team can isolate exactly which agent produced the bad output. With 1 monolithic model, the failure is opaque and debugging is nearly impossible under time pressure.
- Offline resilience: Agent 3 (Evacuation Decision) is rule-based in v1 and runs entirely on-device without internet. Agents 1 and 4 fall back to text. A monolithic cloud model fails completely when connectivity drops during a typhoon.

- Cost and latency: 4 smaller specialized models running sequentially are faster and cheaper than 1 large general model. In a flood emergency, response latency directly affects survival decisions.
- Upgradability: Agent 3 can be upgraded from rule-based to ML in Phase 2 without touching any other agent. Each component evolves independently.

Phase 2: ML Upgrade Path for Agent 3

Agent 3 (Evacuation Decision Agent) is rule-based in the 48-hour MVP. Phase 2 replaces the rule-based logic with a trained ML model using historical Marikina River flood event data, barangay-level vulnerability scores from Dulawan et al. (2024), and seasonal rainfall patterns. Because of the multi-agent architecture, this is a drop-in replacement — Agents 1, 2, and 4 are entirely unchanged.

11.6 Known AI Limitations — Documented Transparently

Limitation	Impact	How AGOS Handles It
Google Cloud STT does not have fully trained models for all Philippine languages in MVP	Dialects with limited training data will have lower accuracy	MVP scope: Tagalog + 1 regional language (Cebuano). Others added Phase 2. Text input always available as fallback.
Evacuation Decision Engine v1 is rule-based, not ML	Cannot learn from new flood patterns or adapt to unusual events	Labelled as rule-based v1 in all outputs. Users see data sources behind every recommendation. ML upgrade path documented for Phase 2.
Route Advisor depends on crowd-report density	In low-adoption areas, sparse reports reduce route confidence	Route outputs include a ‘report density indicator’: how many reports are available near the user’s route. Low density triggers a lower-confidence banner.
All 4 agents require internet for full capability	Typhoon connectivity loss degrades Agent 1 and Agent 4	Documented fallbacks: text input, Tagalog/English text output, cached routes. Every critical function has a non-AI, offline-capable backup path.

SECTION 12 — FEASIBILITY ASSESSMENT

12.1 Technical Risk Register

Risk	Severity	Mitigation
PAGASA dependency	**Low**	PAGASA is not used. Google Flood Forecasting API is the sole primary source. This risk is eliminated entirely.
Crowdsource cold-start (no reports at flood start)	**Medium**	Map pre-seeded by Google Flood API threshold triggers + MMDA EFCOS water-level data. Flood Buddy volunteers are designated as first reporters. Cold-start gap is addressed.
Firebase rate limits during demo	**Low**	MVP stays within the Firebase Spark free tier for prototype-scale testing.
Google Maps SDK billing	**Low**	$200/month free credit — more than sufficient for MVP development and testing.
Semaphore SMS setup time	**Medium**	Registration is same-day. Fallback to FCM-only if SMS integration is delayed beyond the 36-hour mark.
Multilingual STT accuracy	**Medium**	Google Cloud Speech-to-Text with Philippine language models. MVP: Tagalog + 1 regional language. Text input fallback is always available.
Background service permission loss	**Medium**	Permission re-prompt modal on every app open. The user cannot silently lose alert access.
Route computation unavailable offline	**Medium**	Last 3 known-safe routes cached in AsyncStorage. 'Last known route' banner shown when offline. Community road reports sync on restore.
Donation fraud / informal settler exclusion	**Medium**	Address matching + barangay co-verification + household registration code + PhilSys ID option. Three-layer verification system.
Team capacity in 48 hours	**High**	Strictly enforce the MVP scope. Multilingual STT partial (Tagalog + 1). Regeneration Dashboard, donation, and PhilAWARE integration deferred to Phase 2.
Photo upload on slow data	**Medium**	Compress to a max of 500 KB. Text-only submission if upload fails. Queue retries on connectivity restore.
LGU adoption of Admin Portal	**Medium**	Free tier for small barangays. Demo-ready portal with pre-loaded Marikina/Malabon/Navotas seed data. PDRF partnership for LGU onboarding support.

Feasibility Verdict

The 2-day MVP is achievable with clear scope discipline. A functional, demonstrable MVP with Google Flood Forecasting API as the sole real-time data source, MMDA EFCOS flood map bootstrapping, community depth reporting, Community-Informed Route Advisor with offline fallback, offline emergency instructions, multilingual STT (Tagalog + 1 regional), Household Preparedness Kit, rule-based Evacuation Decision Engine, Trash & Pollution Reporter with full lifecycle + Barangay Pollution Score, Google Places resource map, Digital Inclusion Bridge (Flood Buddy + bulk SMS), background service with permission re-prompt, and SMS fallback is achievable within 48 hours by a 3-person team. All core services have free tiers sufficient for MVP. This project directly contributes to the UNDRR Early Warnings for All 2027 target.

SECTION 13 — REFERENCES

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